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Education

International Institute of Information Technology, Bangalore

Jul 2023 – Jul 2028

Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering

CGPA: 3.51/4.00

Experience

Undergraduate Research Affiliate – Multimodal Stress Detection

Jan 2026 – May 2026

Samsung Lab, IIITB | Supervisor: Dr. Sakshi Arora

Bangalore, India

- Designed a multi-stressor protocol (cognitive, physical, and emotional stress tasks) and recorded synchronized 8-channel **EEG** (500 Hz) and **ECG** from 15 subjects with event-marker tagging for phase-aligned labeling.
- Preprocessed raw signals via bandpass filtering, **ICA**-based artifact removal, and R-peak detection, then extracted a 4,939-window feature set spanning EEG band power (delta/theta/alpha/beta) and HRV metrics (**RMSSD**, **LF/HF**, **SDNN**).
- Trained **SVM** and **CNN-LSTM** classifiers under Leave-One-Subject-Out CV, tuning window length and feature subsets to reduce inter-subject variance.
- Achieved **95.6%** subject-independent stress detection accuracy; feature-importance analysis identified frontal beta power and HRV LF/HF ratio as the top contributing signals.

Medical Report Generation from Medical Images

Jan 2026 – Apr 2026

EHRC, IIITB | Supervisor: Dr. Sushree Sangeeta Behera

Bangalore, India

- Architected a large-scale vision-language model (VLM) captioning pipeline for medical imaging, integrating **Virchow2** and **CONCH** vision encoders via FP16 batched GPU inference.
- Engineered a custom **Q-Former** module bridging frozen visual latents into an LLM's embedding space, applying joint CLIP-style contrastive and autoregressive losses for conditional generation.
- Finetuned multimodal VLMs on complex image datasets via a two-phase curriculum—projection warmup followed by **LoRA** mid-training—achieving high-quality conditional text generation (**0.49 ROUGE-1**).
- Developed a scalable evaluation harness to benchmark zero-shot multimodal understanding, assessing prompt-based predictions against ground truth to systematically identify model quality gaps.

Publications

Personalized Multimodal Stress Detection from Consumer Wearables using ECG and Research-Grade EEG

Under Review

Co-Author | Springer Nature Computer Science

Vehicle Detection in Adverse Weather Conditions

2024

Co-Author | Proceedings of the ICDEC VDVWC Challenge, Turkey

Achievements

- Finalist:** NCIIPC AI Grand Challenge - Emerged as one of the top 3 teams (Coders@IIITB) - 100K+ teams. 
- First Prize:** ICDEC'24 International Vehicle Detection Challenge – 1st of 2,000 teams.
- National Runner-Up:** TruthTell Hackathon (WAVES Summit 2025) – Top 2 of 5,600+ global submissions. 
- Attended **Amazon ML Summer School**, one of the top 3000+ students selected out of over 100K applications nationwide.
- 3rd Runner-Up:** Google Gen AI Exchange Hackathon 2025 – 4th among 270,000+ developers. 
- Finalist:** Meta x OpenEnv Hackathon, Agentic AI and Reinforcement Learning Hackathon – 2000+ teams.
- JEE Mains (2023):** Secured a rank of **4439** out of over 11,00,000 applicants.

Projects

RACS: Risk-Aware Cold-Start Recommendation  | Python, PyTorch, Llama 3.1, RecSys

Apr 2026 – May 2026

- Architected a cold-start recommendation framework leveraging **Llama 3.1 (8B)** for contrastive pairwise ranking, harnessing LLM generative priors to resolve preference uncertainty under extreme data sparsity.
- Engineered a Bayesian Semantic Neighborhood Disagreement (SND) module in **PyTorch**, calibrating **CVaR risk thresholds** to dynamically penalize unsafe content and ensure robust exploration for cold-start users.
- Developed an end-to-end offline evaluation harness with leave-one-out protocols to systematically benchmark sequential models (**SASRec**, **EASE**), isolating performance gains via comprehensive ablation studies.
- Achieved a state-of-the-art **0.68 NDCG@5** and 1.0 Hit Rate on MovieLens-1M, significantly outperforming traditional collaborative filtering baselines while rigorously balancing risk-recall trade-offs.

Multi-Objective Optimization for Image Classification 🧠 | *PyTorch, pymoo, BoTorch*

Apr 2026

- Formulated a 3-way **Pareto optimization** problem over a 9-parameter CNN search space to jointly maximize accuracy while minimizing CPU latency and model size.
- Built an automated evaluation loop in **PyTorch** to train sampled architectures and benchmark inference speeds, feeding metrics back to independent optimizers.
- Benchmarked evolutionary search (**pymoo NSGA-II**) against Bayesian optimization (**BoTorch**), with the evolutionary approach yielding a **5.5% larger hypervolume** and denser Pareto front.
- Demonstrated that simple CNN architectures outperformed complex residual models, and identified that CPU cache layouts impact latency more than raw parameter counts.

Domain-Aware Layer-Sensitivity Pruning (DALSP) 🧠 | *PyTorch, Transformers, NLP*

Mar 2026 - Apr 2026

- Architected an **entropy-based adaptive pruning** method for Phi-3.5-mini (3.8B) to improve performance without retraining for domain specific data.
- Attained 20% neuron reduction, 0.56 GB VRAM savings, and 49% throughput gain (11.84 to 17.63 tokens/s).
- Generated deployable domain-specialized subnetworks (Math, Code, Law) via selective neuron masking, enabling lightweight per-domain inference; Math subnet improved GSM8K accuracy from 42% to 53%.
- Implemented per-layer entropy profiling by hooking activations across all transformer blocks during a calibration pass, then thresholding per-domain to select prune candidates without gradient access.

RAG Prompt Compression via LLM Fine-Tuning 🧠 | *PyTorch, Hugging Face, LLMs*

Feb 2026 - Apr 2026

- Engineered a priority-driven prompt compression algorithm and a type-aware AST tokenizer to optimize context windows for Large Language Models.
- Curated a **471K-pair training corpus** with leakage-safe splitting, demonstrating the ability to build scalable workflows for data curation and preparation.
- Executed mid-training workflows by fine-tuning a copy-augmented **CodeT5** pointer-generator over 70K steps, optimizing for maximal token reduction while preserving semantic intent.
- Designed a robust evaluation harness to benchmark generation quality across multiple compression ratios, identifying prompt-format saturation constraints to guide future training iterations.

Vehicle Detection in Adverse Weather Conditions 🧠 | *PyTorch, CycleGAN, YOLOv10, Optuna*

2024

- Developed an **image-to-image editing** pipeline leveraging **CycleGAN** for day-to-night style transfer, evaluating conditional generation capabilities for large-scale synthetic data augmentation.
- Modified neural architectures in **PyTorch** by integrating custom **Squeeze-and-Excitation attention blocks** at the module level, improving spatial feature extraction and attention mechanisms.
- Orchestrated mid-training workflows using **Optuna** for automated hyperparameter sweeps, refining augmentation policies to optimize generalization and rapid model convergence.
- Analyzed failure modes in synthetic image overlays, isolating background diversity as the primary bottleneck to guide scalable data curation for future multimodal training pipelines.

Technical Skills

Languages: Python, C++, SQL, TypeScript, Java, C

AI/ML & Generative AI: PyTorch, Vision-Language Models (VLMs), Conditional Generation, Multimodal Deep Learning, Hugging Face Transformers, Neural Networks

Distributed Systems & DevOps: Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions), High-Throughput Data Pipelines, Kafka, Git, Linux

Technologies & Frameworks: FastAPI, LangChain, CrewAI, MCP, Node.js

Databases: PostgreSQL, MySQL (SQL Systems), MongoDB, TimescaleDB (NoSQL/Time-Series)

Cloud: Amazon Web Services (AWS), Google Cloud Platform (GCP)

Certifications: Google Cloud Platform Certification for Data Science, AI and ML. 📄

Relevant Coursework

Data Structures & Algorithms, Object Oriented Programming (C++ & Java), Operating Systems, Computer Networks, Database Systems, Recommendation Systems, Natural Language Processing, Software Engineering, Machine Learning, Multi-Objective Machine Learning

Extracurricular

- ICP Crewsphere IIITB Chapter (Core Member):** Orchestrated technical sessions and campus-wide hackathons, training and mentoring over 100 students on the ICP Crewsphere platform to foster decentralized application development.
- Zense Developer's Club (Editorial Team):** Curated content and ran the editorial sections for our university's official tech club, working to keep the student developer community informed and engaged.